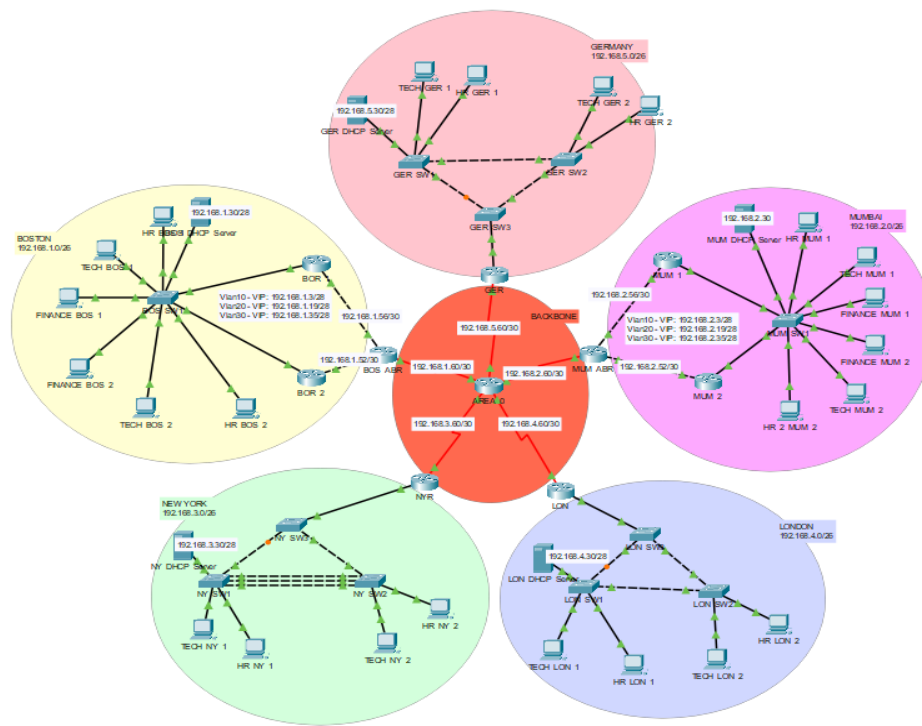


ENTERPRISE NETWORK DESIGN

Network Topology:

This enterprise network is divided into 5 locations assume the locations as Boston, Mumbai, New York, London and Germany. There are two headquarters Boston and Mumbai and other regional offices in New York, London, Germany. Each of these offices is further divided into different departments. Namely Finance, Technical and Human Resources. Only HQ's have the finance department. All these offices intercommunicate using OSPF as the routing protocol. Below is a high-level diagram of the entire network.



This design uses the following hardware devices:

Routers:

- 1) Cisco ISR4321
- 2) Cisco ISR2811

Switches:

- 1) Cisco C2960

Individual location and department network details:

Boston and Mumbai:

- 1) Both Boston and Mumbai (Headquarters) use Star topology to connect all the pc's from each department to a switch.
- 2) The switch is connected to 2 ISR4321 routers. Two routers are used to provide redundancy using HSRP.
- 3) All the devices are configured with Rapid PVST as a spanning tree protocol.
- 4) **Boston** office uses the **192.168.1.0 / 26** subnet to its entire location, where **Mumbai** office used **192.168.2.0 / 26** subnet to its location.
- 5) Both these head offices have Finance, Technical, and Human Resources departments assigned in different VLANs.
 - i) HR – Vlan 10
 - ii) Tech – Vlan 20
 - iii) Finance – Vlan 30

New York, London, Germany:

- 1) These regional offices use Star topology to connect to switches and all the switches are interconnected.
- 2) Rapid PVST is used as a spanning tree protocol to avoid broadcast storms.
- 3) These offices use the below subnets.

New York: 192.168.3.0 / 26

London: 192.168.4.0 / 26

Germany: 192.168.5.0 / 26

- 4) All these subnets are further divided within the location to assign IP's to various departments.
- 5) All the regional offices have Technical and Human Resource Departments in different VLANs.
 - i) HR – Vlan 10
 - ii) Tech – Vlan 20

Assignment of IP Address:

1) Boston: (192.168.1.0 / 26)

Network address: 192.168.1.0

Subnet mask: 255.255.255.192

Wildcard mask: 0.0.0.63

Broadcast address: 192.168.1.63

Usable range: 192.168.1.1 to 192.168.1.62

DHCP/DNS Server: 192.168.1.30/28

PC's:

Department	Vlan	IP ranges
HR	10	192.168.1.4/28 – 192.168.1.14/28
TECH	20	192.168.1.20/28 – 192.168.1.30/28
FINANCE	30	192.168.1.36/28 – 192.168.1.46/28

Routers:

Name	Vlan	IP Address	Virtual IP
BOR1 (g0/0/0.10)	10	192.168.1.1/28	192.168.1.3/28
BOR1 (g0/0/0.20)	20	192.168.1.17/28	192.168.1.19/28
BOR1 (g0/0/0.30)	30	192.168.1.33/28	192.168.1.35/28
BOR1 (g0/0/1)		192.168.1.58/30	
BOR2 (g0/0/0.10)	10	192.168.1.2/28	192.168.1.3/28
BOR2 (g0/0/0.20)	20	192.168.1.18/28	192.168.1.19/28
BOR2 (g0/0/0.30)	30	192.168.1.34/28	192.168.1.35/28
BOR2 (g0/0/1)		192.168.1.54/30	
BOS_ABR (g0/0/0)		192.168.1.57/30	
BOS_ABR (g0/0/1)		192.168.1.53/30	
BOS_ABR (s0/1/0)		192.168.1.62/30	

2) Mumbai: (192.168.2.0 / 26)

Network address: 192.168.2.0

Subnet mask: 255.255.255.192

Wildcard mask: 0.0.0.63

Broadcast address: 192.168.2.63

Usable range: 192.168.2.1 to 192.168.2.62

DHCP Server: 192.168.2.30/28

DNS Server: 192.168.1.30/28

PC's:

Department	Vlan	IP ranges
HR	10	192.168.2.4/28 – 192.168.2.14/28
TECH	20	192.168.2.20/28 – 192.168.2.30/28
FINANCE	30	192.168.2.36/28 – 192.168.2.46/28

Routers:

Name	Vlan	IP Address	Virtual IP
MUM1 (g0/0/0.10)	10	192.168.2.1/28	192.168.2.3/28
MUM1 (g0/0/0.20)	20	192.168.2.17/28	192.168.2.19/28
MUM1 (g0/0/0.30)	30	192.168.2.33/28	192.168.2.35/28
MUM1 (g0/0/1)		192.168.2.58/30	
MUM2 (g0/0/0.10)	10	192.168.2.2/28	192.168.2.3/28
MUM2 (g0/0/0.20)	20	192.168.2.18/28	192.168.2.19/28
MUM2 (g0/0/0.30)	30	192.168.2.34/28	192.168.2.35/28
MUM2 (g0/0/1)		192.168.2.54/30	
MUM_ABR (g0/0/0)		192.168.2.53/30	
MUM_ABR (g0/0/1)		192.168.2.57/30	
MUM_ABR (s0/1/0)		192.168.2.62/30	

3) New York: (192.168.3.0 / 26)

Network address: 192.168.3.0

Subnet mask: 255.255.255.192

Wildcard mask: 0.0.0.63

Broadcast address: 192.168.3.63

Usable range: 192.168.3.1 to 192.168.3.62

DHCP Server: 192.168.3.30/28

DNS Server: 192.168.1.30/28

PC's:

Department	Vlan	IP ranges
HR	10	192.168.3.2/28 – 192.168.3.12/28
TECH	20	192.168.3.18/28 – 192.168.3.28/28

Routers:

Name	Vlan	IP Address
NYR (g0/0/0.10)	10	192.168.3.1/28
NYR (g0/0/0.20)	20	192.168.3.17/28
NYR (s0/0/0)		192.168.3.62/30

4) London: (192.168.4.0 / 26)

Network address: 192.168.4.0

Subnet mask: 255.255.255.192

Wildcard mask: 0.0.0.63

Broadcast address: 192.168.4.63

Usable range: 192.168.4.1 to 192.168.4.62

DHCP Server: 192.168.4.30/28

DNS Server: 192.168.1.30/28

PC's:

Department	Vlan	IP ranges
HR	10	192.168.4.2/28 – 192.168.4.12/28
TECH	20	192.168.4.18/28 – 192.168.4.28/28

Routers:

Name	Vlan	IP Address
LON (g0/0/0.10)	10	192.168.4.1/28
LON (g0/0/0.20)	20	192.168.4.17/28
LON (s0/0/0)		192.168.4.62/30

5) Germany: (192.168.5.0 / 26)

Network address: 192.168.5.0

Subnet mask: 255.255.255.192

Wildcard mask: 0.0.0.63

Broadcast address: 192.168.5.63

Usable range: 192.168.5.1 to 192.168.5.62

DHCP Server: 192.168.5.30/28

DNS Server: 192.168.1.30/28

PC's:

Department	Vlan	IP ranges
HR	10	192.168.5.2/28 – 192.168.5.12/28
TECH	20	192.168.5.18/28 – 192.168.5.28/28

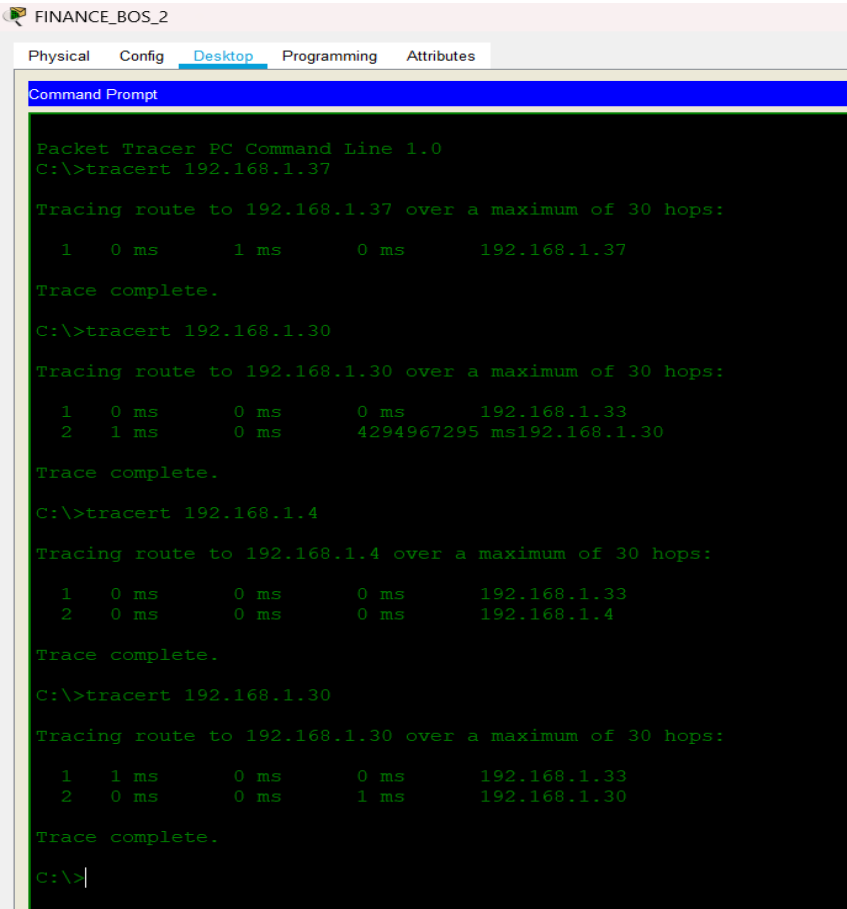
Routers:

Name	Vlan	IP Address
NYR (g0/0/0.10)	10	192.168.5.1/28
NYR (g0/0/0.20)	20	192.168.5.17/28
NYR (s0/0/0)		192.168.5.62/30

TESTS and OUTPUTS:

VLANs:

- 1) VLAN can be tested using trace route command, this command gives the path taken by a packet to reach its destination. In the below Screen shot the packet directly reaches the destination host within same VLAN and reaches other VLANs through router on stick taking 2 hops.



```
FINANCE_BOS_2
Physical Config Desktop Programming Attributes
Command Prompt
Packet Tracer PC Command Line 1.0
C:\>tracert 192.168.1.37

Tracing route to 192.168.1.37 over a maximum of 30 hops:

  1  0 ms      1 ms      0 ms      192.168.1.37
Trace complete.

C:\>tracert 192.168.1.30

Tracing route to 192.168.1.30 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.1.33
  2  1 ms      0 ms     4294967295 ms 192.168.1.30
Trace complete.

C:\>tracert 192.168.1.4

Tracing route to 192.168.1.4 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.1.33
  2  0 ms      0 ms      0 ms      192.168.1.4
Trace complete.

C:\>tracert 192.168.1.30

Tracing route to 192.168.1.30 over a maximum of 30 hops:

  1  1 ms      0 ms      0 ms      192.168.1.33
  2  0 ms      0 ms      1 ms      192.168.1.30
Trace complete.

C:\>|
```

Hops taken to reach different destinations from finance

Routing Protocols:

- 1) In this design **OSPF** is used as the major routing protocol and **EIGRP** is used to route between headquarters via a VPN tunnel.
- 2) Show ip route command gives all the routes learned by the router. Below are the routes learned by the Boston area boundary router.

```
BOS_ABR
Physical Config CLI Attributes
IOS Comm

BOR_ABR#sh ip ro
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
D   1.0.0.0/8 [90/2809856] via 192.168.1.61, 00:07:00, Serial0/1/0
C   1.1.1.1/32 is directly connected, Loopback1
D   2.0.0.0/32 is subnetted, 1 subnets
D   2.2.2.2/32 [90/2809856] via 192.168.1.61, 00:07:00, Serial0/1/0
O   192.168.1.0/24 is variably subnetted, 9 subnets, 3 masks
O   192.168.1.0/28 [110/2] via 192.168.1.58, 00:06:53, GigabitEthernet0/0/0
O   [110/2] via 192.168.1.54, 00:06:53, GigabitEthernet0/0/1
O   192.168.1.16/28 [110/2] via 192.168.1.58, 00:06:53, GigabitEthernet0/0/0
O   [110/2] via 192.168.1.54, 00:06:53, GigabitEthernet0/0/1
O   192.168.1.32/28 [110/2] via 192.168.1.58, 00:06:53, GigabitEthernet0/0/0
O   [110/2] via 192.168.1.54, 00:06:53, GigabitEthernet0/0/1
C   192.168.1.52/30 is directly connected, GigabitEthernet0/0/1
L   192.168.1.53/32 is directly connected, GigabitEthernet0/0/1
C   192.168.1.56/30 is directly connected, GigabitEthernet0/0/0
L   192.168.1.57/32 is directly connected, GigabitEthernet0/0/0
C   192.168.1.60/30 is directly connected, Serial0/1/0
L   192.168.1.62/32 is directly connected, Serial0/1/0
D   192.168.2.0/24 is variably subnetted, 7 subnets, 3 masks
O   192.168.2.0/24 [90/2681856] via 192.168.1.61, 00:07:00, Serial0/1/0
O IA 192.168.2.0/28 [110/130] via 192.168.1.61, 00:06:38, Serial0/1/0
O IA 192.168.2.16/28 [110/130] via 192.168.1.61, 00:06:38, Serial0/1/0
O IA 192.168.2.32/28 [110/130] via 192.168.1.61, 00:06:38, Serial0/1/0
O IA 192.168.2.52/30 [110/129] via 192.168.1.61, 00:06:38, Serial0/1/0
O IA 192.168.2.56/30 [110/129] via 192.168.1.61, 00:06:38, Serial0/1/0
O   192.168.2.60/30 [110/128] via 192.168.1.61, 00:06:38, Serial0/1/0
D   192.168.3.0/24 is variably subnetted, 3 subnets, 2 masks
O IA 192.168.3.0/28 [110/129] via 192.168.1.61, 00:06:38, Serial0/1/0
O IA 192.168.3.16/28 [110/129] via 192.168.1.61, 00:06:38, Serial0/1/0
O   192.168.3.60/30 [110/128] via 192.168.1.61, 00:06:38, Serial0/1/0
D   192.168.4.0/24 is variably subnetted, 3 subnets, 2 masks
O IA 192.168.4.0/28 [110/129] via 192.168.1.61, 00:06:38, Serial0/1/0
O IA 192.168.4.16/28 [110/129] via 192.168.1.61, 00:06:38, Serial0/1/0
O   192.168.4.60/30 [110/128] via 192.168.1.61, 00:06:38, Serial0/1/0
D   192.168.5.0/24 is variably subnetted, 3 subnets, 2 masks
O IA 192.168.5.0/28 [110/129] via 192.168.1.61, 00:06:38, Serial0/1/0
O IA 192.168.5.16/28 [110/129] via 192.168.1.61, 00:06:38, Serial0/1/0
O   192.168.5.60/30 [110/128] via 192.168.1.61, 00:06:38, Serial0/1/0
--More--
```

Routes learned by the BOS_ABR

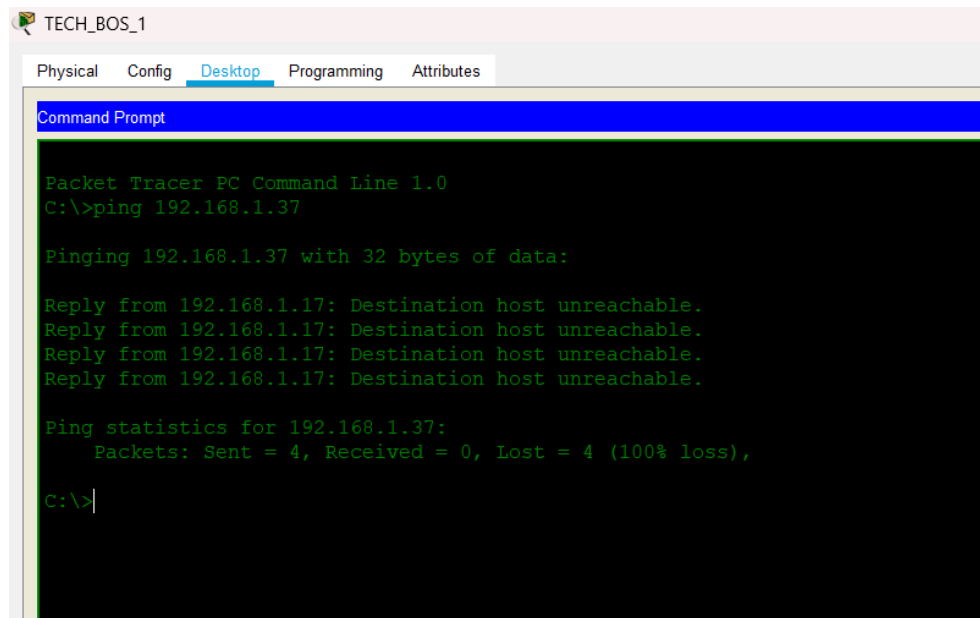
```
BOS_ABR
Physical Config CLI Attributes

L   192.168.1.62/32 is directly connected, Serial0/1/0
D   192.168.2.0/24 is variably subnetted, 7 subnets, 3 masks
O IA 192.168.2.0/28 [110/130] via 192.168.1.61, 00:06:38, Serial0/1/0
O IA 192.168.2.16/28 [110/130] via 192.168.1.61, 00:06:38, Serial0/1/0
O IA 192.168.2.32/28 [110/130] via 192.168.1.61, 00:06:38, Serial0/1/0
O IA 192.168.2.52/30 [110/129] via 192.168.1.61, 00:06:38, Serial0/1/0
O IA 192.168.2.56/30 [110/129] via 192.168.1.61, 00:06:38, Serial0/1/0
O   192.168.2.60/30 [110/128] via 192.168.1.61, 00:06:38, Serial0/1/0
D   192.168.3.0/24 is variably subnetted, 3 subnets, 2 masks
O IA 192.168.3.0/28 [110/129] via 192.168.1.61, 00:06:38, Serial0/1/0
O IA 192.168.3.16/28 [110/129] via 192.168.1.61, 00:06:38, Serial0/1/0
O   192.168.3.60/30 [110/128] via 192.168.1.61, 00:06:38, Serial0/1/0
D   192.168.4.0/24 is variably subnetted, 3 subnets, 2 masks
O IA 192.168.4.0/28 [110/129] via 192.168.1.61, 00:06:38, Serial0/1/0
O IA 192.168.4.16/28 [110/129] via 192.168.1.61, 00:06:38, Serial0/1/0
O   192.168.4.60/30 [110/128] via 192.168.1.61, 00:06:38, Serial0/1/0
D   192.168.5.0/24 is variably subnetted, 3 subnets, 2 masks
O IA 192.168.5.0/28 [110/129] via 192.168.1.61, 00:06:38, Serial0/1/0
O IA 192.168.5.16/28 [110/129] via 192.168.1.61, 00:06:38, Serial0/1/0
O   192.168.5.60/30 [110/128] via 192.168.1.61, 00:06:38, Serial0/1/0
```

Routes learned by the BOS_ABR

Security:

- 1) In this design ACLs, VPN, Mac limit is used as the security measures to protect access and data into the network.
- 2) ACLs are used to prevent unauthorized access to finance by other departments (TECH and HR).
- 3) ACLs are tested by pinging finance from other departments.
- 4) VPN is used to transfer data securely by encrypting between both the HQs.
- 5) Mac-flooding attack is defended by limiting the maximum devices on a switch port.
- 6) Below are the Screen shots of testing all the security measures taken.



```
TECH_BOS_1
Physical Config Desktop Programming Attributes
Command Prompt
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.37

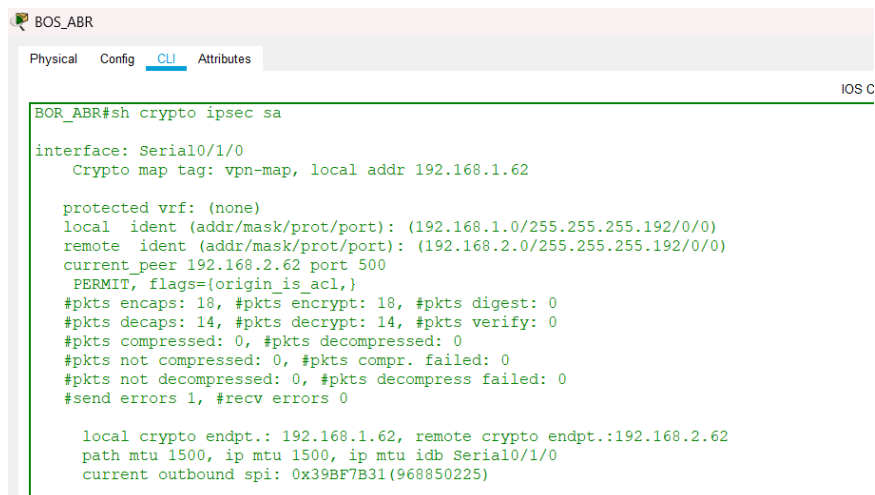
Pinging 192.168.1.37 with 32 bytes of data:

Reply from 192.168.1.17: Destination host unreachable.
Reply from 192.168.1.17: Destination host unreachable.
Reply from 192.168.1.17: Destination host unreachable.
Reply from 192.168.1.17: Destination host unreachable.

Ping statistics for 192.168.1.37:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Ping from Tech to Finance is blocked by ACL.



```
BOS_ABR
Physical Config CLI Attributes
IOS C
BOS_ABR#sh crypto ipsec sa

interface: Serial0/1/0
  Crypto map tag: vpn-map, local addr 192.168.1.62

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.192/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.255.192/0/0)
current_peer 192.168.2.62 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 18, #pkts encrypt: 18, #pkts digest: 0
#pkts decaps: 14, #pkts decrypt: 14, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

local crypto endpt.: 192.168.1.62, remote crypto endpt.:192.168.2.62
path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/0
current outbound spi: 0x39BF7B31(968850225)
```

Packets are being encrypted and decrypted at BOS_ABR

```

MUM_ABR#sh cr ipsec sa

interface: Serial0/1/0
  Crypto map tag: vpn, local addr 192.168.2.62

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.2.0/255.255.255.192/0/0)
remote ident (addr/mask/prot/port): (192.168.1.0/255.255.255.192/0/0)
current peer 192.168.1.62 port 500
  PERMIT, flags=(origin_is_acl,)
  #pkts encaps: 14, #pkts encrypt: 14, #pkts digest: 0
  #pkts decaps: 18, #pkts decrypt: 18, #pkts verify: 0
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

local crypto endpt.: 192.168.2.62, remote crypto endpt.: 192.168.1.62
path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/0
current outbound spi: 0x15877585(361198981)

```

```

NY_SW3
Physical Config CLI Attributes

!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
  switchport trunk native vlan 1001
  switchport trunk allowed vlan 10,20
  switchport mode trunk
  switchport port-security maximum 30
!
interface FastEthernet0/2
  switchport trunk native vlan 1001
  switchport trunk allowed vlan 10,20
  switchport mode trunk
  switchport port-security maximum 30
!
interface FastEthernet0/3
  switchport trunk native vlan 1001
  switchport trunk allowed vlan 10,20
  switchport mode trunk
  switchport port-security maximum 30
!

```

R

Packets are being encrypted and decrypted at MUM_ABR

No of devices limited to 30

The top screenshot shows the configuration of NY_SW3 in the CLI. The configuration includes spanning-tree mode rapid-pvst, spanning-tree extend system-id, and three FastEthernet interfaces (0/1, 0/2, 0/3) configured as trunks with native VLAN 1001 and allowed VLANs 10 and 20. Port security is configured with a maximum of 30 ports.

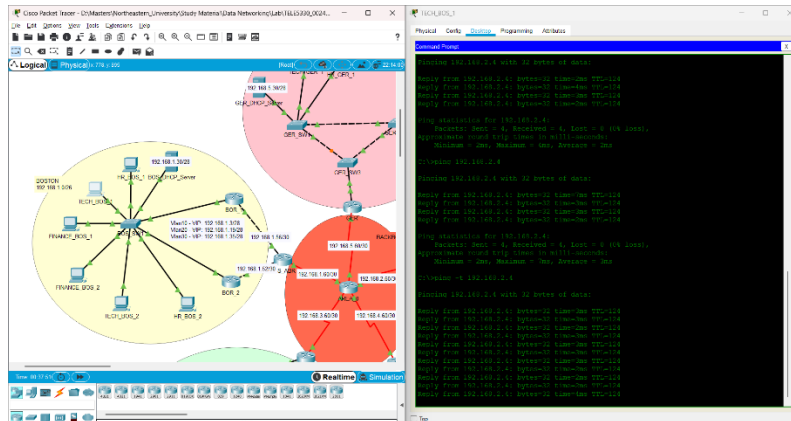
The bottom screenshot shows the configuration of MUM_ABR in the CLI. The configuration includes the crypto map tag 'vpn' on interface Serial0/1/0, with local address 192.168.2.62 and remote address 192.168.1.62. The crypto map is configured with PERMIT, flags=(origin_is_acl,), and the current peer is 192.168.1.62 port 500. The crypto map is configured with the following parameters: #pkts encaps: 14, #pkts encrypt: 14, #pkts digest: 0; #pkts decaps: 18, #pkts decrypt: 18, #pkts verify: 0; #pkts compressed: 0, #pkts decompressed: 0; #pkts not compressed: 0, #pkts compr. failed: 0; #pkts not decompressed: 0, #pkts decompress failed: 0; #send errors 0, #recv errors 0. The local crypto endpt. is 192.168.2.62, the remote crypto endpt. is 192.168.1.62, the path mtu is 1500, the ip mtu is 1500, the ip mtu idb is Serial0/1/0, and the current outbound spi is 0x15877585(361198981).

The middle screenshot shows a packet capture of an ICMP packet. The packet is captured on the interface Serial0/1/0 of MUM_ABR. The packet is an ICMP Echo (ping) packet. The packet is encrypted with ESP. The packet is captured with the following details: Time: 0.000, Source: 192.168.1.62, Destination: 192.168.2.62, Protocol: ICMP, Length: 60, Encapsulated: 40. The packet is captured with the following details: Time: 0.000, Source: 192.168.1.62, Destination: 192.168.2.62, Protocol: ICMP, Length: 60, Encapsulated: 40. The packet is captured with the following details: Time: 0.000, Source: 192.168.1.62, Destination: 192.168.2.62, Protocol: ICMP, Length: 60, Encapsulated: 40.

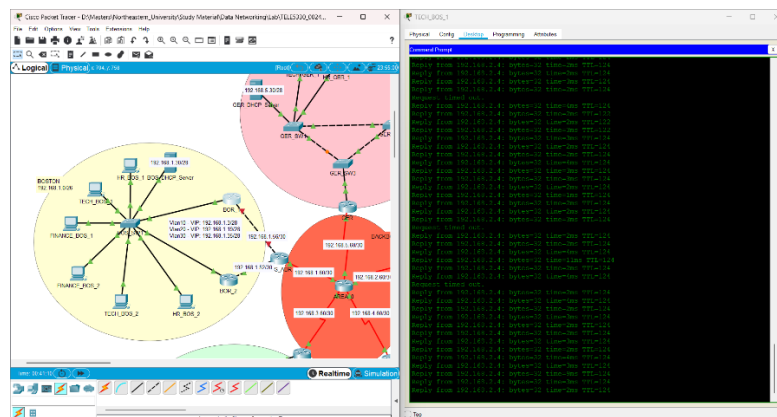
ICMP packets showing the encryption (ESP) header

Redundancy:

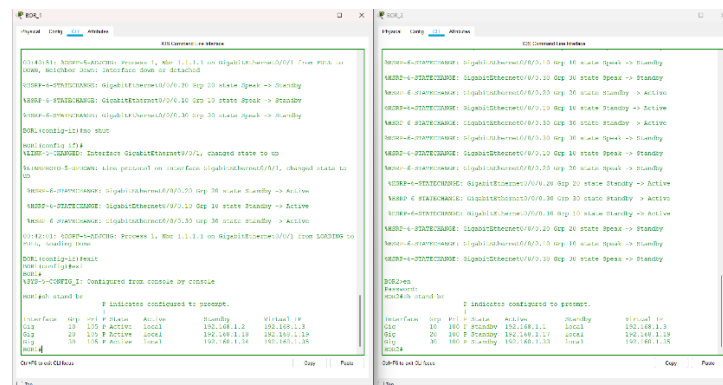
- 1) HSRP and STP are implemented as redundancy protocol in this design.
- 2) HSRP provides redundancy with the help of a backup router which takes over when there is a fault in the active router.
- 3) HSRP is tested by giving a continuous ping and shutting down the port manually and testing the connectivity.



Packets are routed through BOR_1



Link between BOR_1 and BOS_ABR is shut



Active and Standby routers

- 4) STP is used to provide redundancies between switches.
- 5) STP eliminates the ethernet loops by blocking the redundant ports and enables these ports to be active if any active ports malfunction. STP ensures that no loops are formed and avoids broadcast floods. Below are STP configurations.

The following tables represent the STP configurations shown in the screenshots for each switch.

NY_SW2 Configuration

```

NY_SW2#sh spanning-tree
VLAN0010
Spanning tree enabled protocol rstp
Root ID    Priority    32776
Address    0001.C7D6.705C
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Bridge ID   Priority    32776 (priority 32768 sys-id-ext 10)
Address     0001.C7D6.705C
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type
-----
Fa0/2 Desg FWD 19 128.2 P2p
Fa0/4 Desg FWD 19 128.4 P2p
Po1 Desg FWD 8 120.27 Shr
  
```

NY_SW1 Configuration

```

NY_SW1#sh spanning-tree
VLAN0010
Spanning tree enabled protocol rstp
Root ID    Priority    32779
Address    0001.C7D6.705C
Cost       6
Port       27 (FastEthernet1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Bridge ID   Priority    32779 (priority 32768 sys-id-ext 10)
Address     0001.C7D6.705C
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type
-----
Fa0/2 Desg FWD 19 128.2 P2p
Fa0/4 Desg FWD 19 128.4 P2p
Po1 Desg FWD 8 120.27 Shr
  
```

NY_SW3 Configuration

```

NY_SW3#sh spanning-tree
VLAN0010
Spanning tree enabled protocol rstp
Root ID    Priority    32788
Address    0001.C7D6.705C
Cost       19
Port       2 (FastEthernet0/2)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

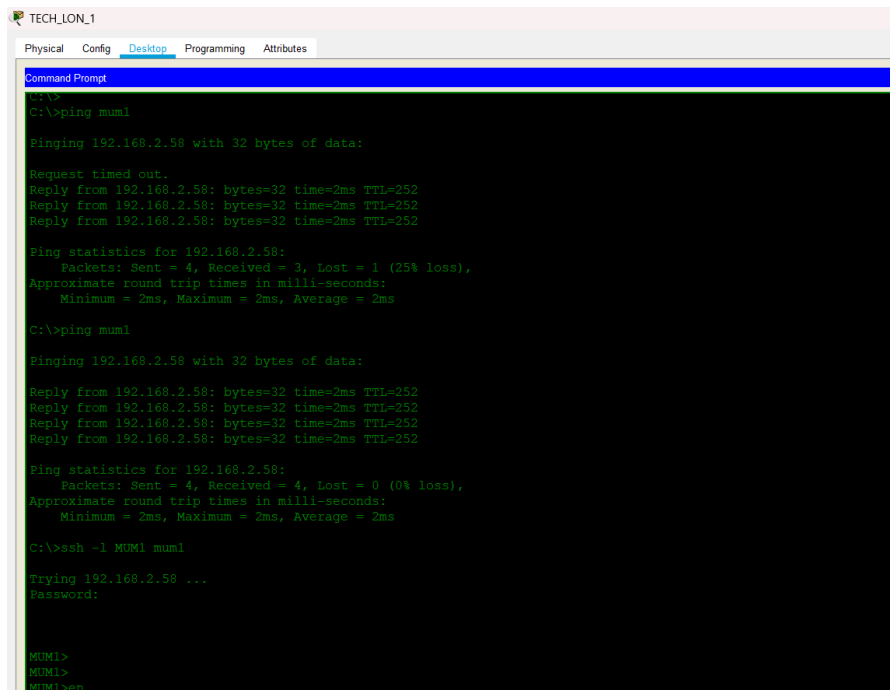
Bridge ID   Priority    32788 (priority 32768 sys-id-ext 20)
Address     0002.4A3E.2BA0
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type
-----
Fa0/1 Desg FWD 19 128.1 P2p
Fa0/2 Desg FWD 19 128.2 P2p
Fa0/3 Desg FWD 19 128.3 P2p
Po1 Desg FWD 8 120.27 Shr
  
```

STP configurations

SSH:

- 1) Secure Shell is used to secure remote login on to a distance host such as routers, servers and other network devices.
- 2) SSH by default uses TCP port 22 for communication.
- 3) In this network all the routers provide SSH access, and these routers can be accessed by the domain names which are translated by a DNS server.



```
TECH_LON_1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping mumi
Pinging 192.168.2.58 with 32 bytes of data:
Request timed out.
Reply from 192.168.2.58: bytes=32 time=2ms TTL=252
Reply from 192.168.2.58: bytes=32 time=2ms TTL=252
Reply from 192.168.2.58: bytes=32 time=2ms TTL=252

Ping statistics for 192.168.2.58:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\>ping mumi
Pinging 192.168.2.58 with 32 bytes of data:
Reply from 192.168.2.58: bytes=32 time=2ms TTL=252
Reply from 192.168.2.58: bytes=32 time=2ms TTL=252
Reply from 192.168.2.58: bytes=32 time=2ms TTL=252
Reply from 192.168.2.58: bytes=32 time=2ms TTL=252

Ping statistics for 192.168.2.58:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

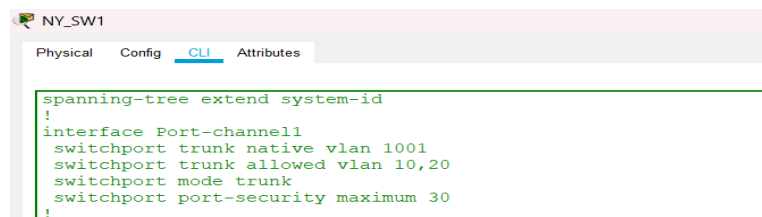
C:\>ssh -l MUMI mumi
Trying 192.168.2.58 ...
Password:

MUMI>
MUMI>
MUMI>exit
```

SSH login on to Mumbai router from London tech

Ether channel:

- 1) Ether channel allows to bundle multiple physical interfaces into a single virtual group channel or port channel.
- 2) It helps to increase bandwidth, load balance and fault tolerance.
- 3) In this project Ether channel is implemented in New York.

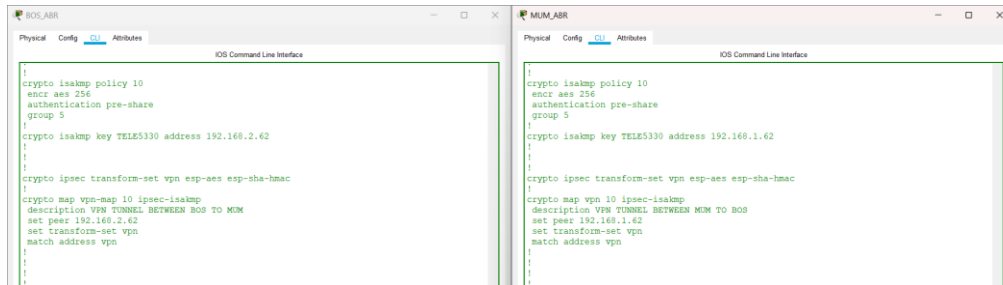


```
NY_SW1
Physical Config CLI Attributes
spanning-tree extend system-id
!
interface Port-channel1
 switchport trunk native vlan 1001
 switchport trunk allowed vlan 10,20
 switchport mode trunk
 switchport port-security maximum 30
!
```

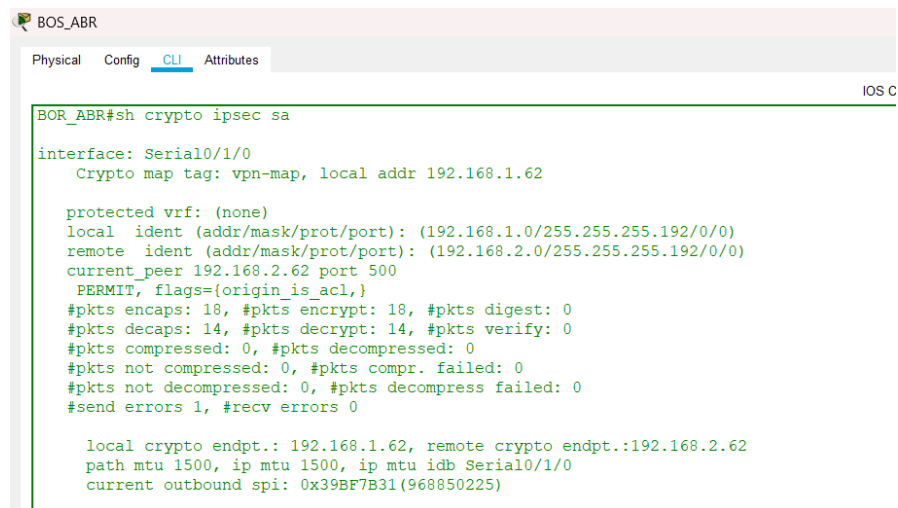
Ether channel interface

VPN:

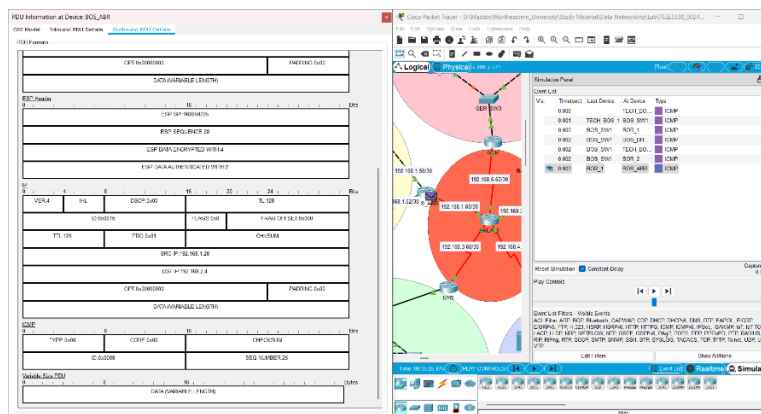
- 1) VPN is configured between Boston ABR and Mumbai ABR using IPSEC, isakmp and pre shared key which is TELE5330.
- 2) ICMP packets are encrypted with ESP header.



VPN configurations



Packet encryption by BOS_ABR



ESP header added to the packet

Concepts Implemented:

1) Subnetting

- i) Subnets
- ii) Network Mask
- iii) Wildcard Mask

2) Virtual LANs

- i) Access ports
- ii) Trunk Ports
- iii) Router on a stick

3) Spanning Tree Protocol

- i) Broadcast storms
- ii) Bridge ID, Port status (Root, Designated, Non-designated ports)
- iii) PVST, MSTP, Rapid STP
- iv) Port fast, BPDU

4) Hot Standby Router Protocol

- i) Gratuitous ARP
- ii) Load balance
- iii) Virtual Ips

5) Dynamic Routing

- i) Open Shortest Path First (OSPF)
- ii) Routing information Protocol (RIP)
- iii) Enhanced Interior Gateway Routing Protocol (EIGRP)
- iv) Routing Metric

6) Access control Lists (ACLs):

- i) Standard ACLs
- ii) Extended ACLs

7) Domain Name System (DNS)

8) Dynamic Host Configuration Protocol (DHCP)

9) Ether Channel

10) Secure Shell Access (SSH)

11) Virtual Private Network (VPN)

Conclusion:

In conclusion this project to design an enterprise design helped me to learn all the above-mentioned concepts both theoretical and practical implementation using cisco packet tracer. The high-level view of the is to design a multi-location, inter-network solution for an organization with various requirements. It helped me understand and gain knowledge of network technologies and protocols. It also made me understand the network hardware, software, various security concerns of a network and a few ways to defend against attacks.

This enterprise network has implemented all the networking concepts such as Virtual Lans, Subnets, Redundancy, Dynamic routing, Security, and optimization of financial resources. Overall, this project helps to develop a sustainable network design solution with all the necessary industry requirements.